

Physics and Modelling of the Ball Balancing Platform Simulator

Model specification, derivations and verification

Ball Balancing Platform project

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Abstract

This document specifies, completely and independently of its implementation, the physical model behind the Ball Balancing Platform (BBP) simulator: the kinematics and dynamics of the three-legged (3-RRS) parallel mechanism that tilts the plate, the hobby-servo actuators, the rolling, slipping, sticking and bouncing of the ball on the moving plate (including its rolling inertia), and the formation of the image seen by the camera that looks up through the transparent plate. Every equation that the simulator integrates is stated here, every parameter is listed with its source, and every simplification is named. The model is verified against classical closed-form results—the $\frac{5}{7}g \sin \beta$ law of rolling, the $\frac{2}{7}\Omega$ orbit of a ball on a turntable, the $\frac{5}{7}v_0$ transition from sliding to rolling, Snell refraction through a slab—and the agreement is reported to machine precision where the theory is exact.

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1 Purpose and scope

The BBP is a teaching platform for computer vision and control. Students write a ball tracker and a controller; the same code runs on the physical platform (on a DFRobot FireBeetle 2 ESP32-S3) and in a simulator they can use at home. For the simulator to be a fair stand-in for the hardware, students must be able to see *what* it assumes, even though the code that implements it is not distributed. This document is that statement: the physics is transparent, the implementation is not.

The simulated system comprises

1. the **mechanism**: a level-able plate carried by three crank-rod legs (sections 2 to 4);
2. the **actuators**: three position-controlled hobby servos driven by PWM (section 5);
3. the **ball**: a rigid sphere in frictional, unilateral contact with the moving plate (section 6);
4. the **camera**: an image sensor below the plate, seeing the ball through it (section 7).

Section 8 describes how the equations are integrated, section 9 the verification, and section 11 what is *not* modelled. The modelling of the student-facing artificial delays and of processor execution time belongs to the software framework and is specified separately.

Conventions. Vectors are bold and, unless stated otherwise, expressed in the world frame. SI units are used throughout; lengths in tables are given in millimetres where that is clearer. A dot denotes a time derivative in the world frame. $[\mathbf{a}]_{\times}$ is the cross-product matrix, $[\mathbf{a}]_{\times}\mathbf{b} = \mathbf{a} \times \mathbf{b}$. For a unit normal $\mathbf{n}$, the tangential part of a vector is $\mathbf{x}_t = \mathbf{x} - (\mathbf{x} \cdot \mathbf{n})\mathbf{n}$. Parameter provenance is tagged CAD (measured from the CAD model), DS (component datasheet), DR (derived) and AS (assumed; to be identified on the real platform).

2 Geometry from the CAD model

All geometric parameters are extracted automatically from the STEP assembly `CAD/FullASSY.step` by a script (`tools/cad/extract_geometry.py`) that identifies the joints by their characteristic surfaces: the crank hub cylinders give the servo axes, the crank-pin holes the lower joints of the rods, and the 8 mm balls the spherical joints under the plate. The values are listed in table 1.

Table 1: Geometry of the mechanism (CAD). The world frame is the CAD frame: z up, origin on the central axis at the height of the plate’s lower face in the as-modelled pose.

Quantity	Symbol	Value
Leg azimuths	β_i	$-150^\circ, -30^\circ, 90^\circ$
Radius of servo (crank) axes	r_A	44.007 mm
Height of servo axes	z_A	-90.934 mm
Crank length (axis to pin)	L_1	50.000 mm
Rod length (pin to sphere centre)	L_2	118.598 mm
Radius of spherical-joint circle	r_B	94.000 mm
Depth of joint centres below plate top	h_s	13.598 mm
Plate radius, thickness	R_p, t_p	100 mm, 5 mm
Ball radius	a	10.000 mm
Camera optical centre	$\mathbf{C}$	$(0, 0, -92.404)$ mm
Crank angle in the CAD pose	θ_{CAD}	-45.004°

Idealisation. In the CAD assembly the mid-section is rotated by 0.47° , so each leg plane passes 0.77 mm beside the central axis. The model uses the design intent: exact 120° symmetry with leg planes through the axis. The forward kinematics of the idealised model reproduces the CAD plate height in the CAD pose to within $2.6\ \mu\text{m}$ (section 9), which bounds the effect of this idealisation.

Operating pose. The CAD shows the cranks 45° below horizontal. The simulator’s neutral pose has the cranks horizontal ($\theta_i = 0$, servo pulse $1500\ \mu\text{s}$), where each rod is vertical and perpendicular to its crank—the best force transmission. The plate top is then at $h_0 = 41.263\ \text{mm}$, 133.67 mm above the camera. Both choices are parameters.

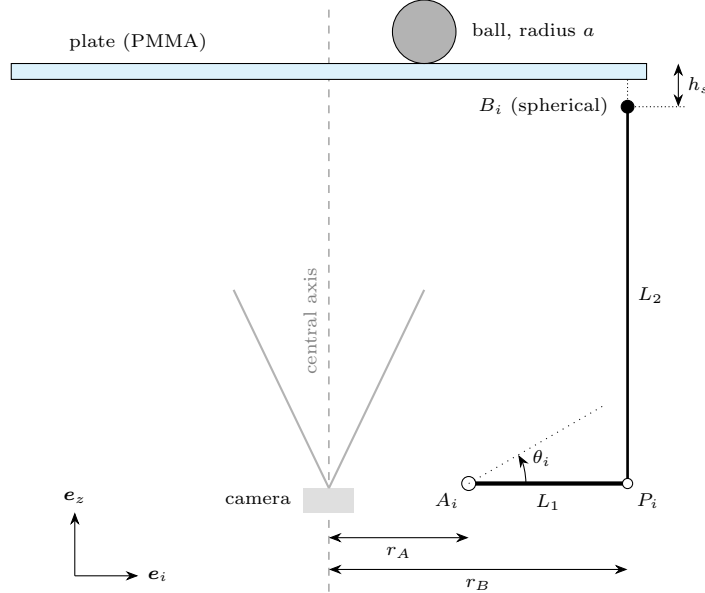


Figure 1: One leg in its vertical plane, drawn to scale in the neutral pose ($\theta_i = 0$). The servo turns the crank A_iP_i about an axis along $\mathbf{t}_i$; the rod P_iB_i is hinged at P_i about a parallel axis and ends in a spherical joint B_i fixed to the plate. The camera looks up through the transparent plate.

3 Kinematics of the 3-RRS mechanism

Each leg is an actuated revolute joint (servo), a passive revolute joint (crank pin) and a spherical joint (under the plate): a 3-RRS parallel mechanism with three degrees of freedom, kinematically equivalent to the classical 3-RPS manipulator of Lee and Shah [23] in that every spherical joint is confined to a fixed vertical plane [25, 33].

3.1 Frames and loop-closure constraints

Leg i lies in the vertical plane through the central axis at azimuth β_i , spanned by $\mathbf{e}_i = (\cos \beta_i, \sin \beta_i, 0)^T$ and $\mathbf{e}_z$; its normal is $\mathbf{t}_i = \mathbf{e}_z \times \mathbf{e}_i$. The servo axis passes through $\mathbf{A}_i = r_A \mathbf{e}_i + z_A \mathbf{e}_z$ along $\mathbf{t}_i$, and the crank pin is at

$$\mathbf{P}_i(\theta_i) = \mathbf{A}_i + L_1 (\cos \theta_i \mathbf{e}_i + \sin \theta_i \mathbf{e}_z), \quad (1)$$

with θ_i measured from the outward horizontal, positive when the pin rises.

The plate frame P has its origin $\mathbf{p}$ at the centre of the plate’s *top* face and its z axis along the plate normal $\mathbf{n} = \mathbf{R}\mathbf{e}_z$; $\mathbf{R}$ maps plate to world coordinates. The spherical joints sit at plate coordinates $\mathbf{b}_i = (r_B \cos \beta_i, r_B \sin \beta_i, -h_s)^T$, i.e. at $\mathbf{B}_i = \mathbf{p} + \mathbf{R}\mathbf{b}_i$. The plate pose must satisfy six constraints,

$$g_{2i-1} = \frac{\|\mathbf{B}_i - \mathbf{P}_i(\theta_i)\|^2 - L_2^2}{2L_2} = 0 \quad (\text{rod length}), \quad (2)$$

$$g_{2i} = \mathbf{t}_i \cdot \mathbf{B}_i = 0 \quad (\text{joint in the leg plane}), \quad (3)$$

for $i = 1, 2, 3$. The scaling of (2) gives both rows units of length, which keeps the Newton iteration below well conditioned.

3.2 Orientation: tilt-and-torsion angles

Following Bonev et al. [5], the plate orientation is written as

$$\mathbf{R} = \mathbf{R}_z(\phi) \mathbf{R}_y(\vartheta) \mathbf{R}_z(\sigma - \phi), \quad (4)$$

where ϑ is the *tilt* (angle between plate normal and vertical), ϕ the *azimuth* of the tilt and σ the *torsion*. The plate normal depends only on (ϕ, ϑ) : $\mathbf{n} = (\sin \vartheta \cos \phi, \sin \vartheta \sin \phi, \cos \vartheta)^T$. For the student interface the tilt is specified by two *surface slope angles* (s_x, s_y) —the angles the plate surface makes with the horizontal along the world x and y directions—through $\mathbf{n} \propto (-\tan s_x, -\tan s_y, 1)^T$. Unlike a pair of Euler angles this definition does not depend on a rotation order.

3.3 Zero torsion and parasitic translation

Let $\mathbf{p}_J = \mathbf{p} - h_s \mathbf{n}$ be the centre of the joint circle. Substituting $\mathbf{B}_i = \mathbf{p}_J + r_B \mathbf{R}(\cos \beta_i, \sin \beta_i, 0)^T$ with $\sigma = 0$ into (3) and writing $\psi_i = \beta_i - \phi$ gives, after elementary trigonometry,

$$-\sin \beta_i p_{J,x} + \cos \beta_i p_{J,y} = -\frac{r_B}{2} (1 - \cos \vartheta) \sin 2\psi_i. \quad (5)$$

For legs spaced by 120° , $2\beta_i \equiv 3\beta_1 - \beta_i \pmod{2\pi}$, so the right-hand side becomes $-\frac{r_B}{2} (1 - \cos \vartheta) \sin(\gamma - \beta_i)$ with $\gamma = 3\beta_1 - 2\phi$: a first harmonic in β_i , exactly like the left-hand side. All three equations are therefore satisfied by

$$\sigma = 0, \quad \begin{pmatrix} p_{J,x} \\ p_{J,y} \end{pmatrix} = -\frac{r_B}{2} (1 - \cos \vartheta) \begin{pmatrix} \cos \gamma \\ \sin \gamma \end{pmatrix}. \quad (6)$$

This mechanism is thus a *zero-torsion* mechanism in the sense of Bonev et al. [5]: it never yaws, and it translates sideways by $\frac{r_B}{2} (1 - \cos \vartheta)$ —0.71 mm at 10° of tilt—in a direction that turns at twice the tilt azimuth. The same result is known for the 3-PRS and 3-RPS families [8]. Heave p_z is free.

3.4 Inverse kinematics

Given the desired normal $\mathbf{n}$ and plate-top height h , the pose follows from (6): $\mathbf{R}$ from (ϕ, ϑ) with $\sigma = 0$, p_x, p_y from $\mathbf{p} = \mathbf{p}_J + h_s \mathbf{n}$, and $p_z = h$. (The implementation refines (p_x, p_y, σ) with a few Newton steps on (3), which makes it exact for any leg layout and changes nothing for the symmetric one.) Each crank angle then follows in closed form. With the joint centre expressed in leg-plane coordinates relative to the servo axis, $u_i = \mathbf{e}_i \cdot \mathbf{B}_i - r_A$ and $v_i = B_{i,z} - z_A$, the rod-length condition $(u_i - L_1 \cos \theta_i)^2 + (v_i - L_1 \sin \theta_i)^2 = L_2^2$ is solved by

$$\theta_i = \text{atan2}(v_i, u_i) - \arccos \frac{K_i}{\rho_i}, \quad K_i = \frac{u_i^2 + v_i^2 + L_1^2 - L_2^2}{2L_1}, \quad \rho_i = \sqrt{u_i^2 + v_i^2}, \quad (7)$$

taking the branch of the assembled mechanism (crank pin outside and below the line $A_i B_i$). The pose is reachable if and only if $|K_i| \leq \rho_i$ for all legs.

3.5 Forward kinematics

Given $\boldsymbol{\theta}$, the pose is found by Newton's method on the six constraints (2)–(3). The unknown is updated as a twist $\delta \mathbf{x} = (\delta \mathbf{p}, \delta \boldsymbol{\varphi})$: $\mathbf{p} \leftarrow \mathbf{p} + \delta \mathbf{p}$, $\mathbf{R} \leftarrow \exp([\delta \boldsymbol{\varphi}]_\times) \mathbf{R}$ (Rodrigues' formula), followed by re-orthonormalisation. Since $\delta \mathbf{B}_i = \delta \mathbf{p} + \delta \boldsymbol{\varphi} \times \mathbf{R} \mathbf{b}_i$, the rows of the constraint Jacobian $\boldsymbol{\Phi}_x$ are

$$\frac{\partial g_{2i-1}}{\partial \delta \mathbf{x}} = [\mathbf{d}_i^T, (\mathbf{R} \mathbf{b}_i \times \mathbf{d}_i)^T], \quad \frac{\partial g_{2i}}{\partial \delta \mathbf{x}} = [\mathbf{t}_i^T, (\mathbf{R} \mathbf{b}_i \times \mathbf{t}_i)^T], \quad \mathbf{d}_i = \frac{\mathbf{B}_i - \mathbf{P}_i}{L_2}. \quad (8)$$

In the simulator the iteration is warm-started from the previous time step and converges in one or two iterations.

3.6 Velocity Jacobian and singularities

Differentiating the constraints gives $\boldsymbol{\Phi}_x \boldsymbol{\xi} + \boldsymbol{\Phi}_\theta \dot{\boldsymbol{\theta}} = \mathbf{0}$ for the plate twist $\boldsymbol{\xi} = (\mathbf{v}_p, \boldsymbol{\omega}_p)$, where $\mathbf{v}_p$ is the velocity of $\mathbf{p}$ and $\boldsymbol{\omega}_p$ the angular velocity. $\boldsymbol{\Phi}_\theta$ has a single non-zero entry per leg, $\partial g_{2i-1} / \partial \theta_i = -\mathbf{d}_i \cdot \mathbf{P}'_i(\theta_i)$. Hence

$$\boldsymbol{\xi} = \mathbf{J} \dot{\boldsymbol{\theta}}, \quad \mathbf{J} = -\boldsymbol{\Phi}_x^{-1} \boldsymbol{\Phi}_\theta \in \mathbb{R}^{6 \times 3}. \quad (9)$$

The mechanism is singular when $\boldsymbol{\Phi}_\theta$ loses rank (a crank aligned with its rod, *type I*) or when $\boldsymbol{\Phi}_x$ does (*type II*, the plate gains a degree of freedom), following Gosselin and Angeles [15]. Neither occurs in the working range; the simulator stops with an error if one is reached.

4 Mechanism dynamics

4.1 Inertial model

- **Plate:** mass, centre of mass and inertia tensor from the CAD solid scaled by the density of PMMA, $\rho = 1190 \text{ kg/m}^3$ (AS material): $m_p = 0.188 \text{ kg}$, $I_{xx} = I_{yy} = 4.73 \times 10^{-4}$, $I_{zz} = 9.44 \times 10^{-4} \text{ kg m}^2$, centre of mass 2.53 mm below the top face.
- **Rods:** each rod (10.6 g, solid PLA, AS) is replaced by two point masses, half at the spherical joint (added to the plate) and half at the crank pin (added to the crank). This two-point lumping preserves mass and centre of mass but not the rod's rotational inertia, which is negligible here.
- **Cranks:** rotate about fixed axes; inertia about the axis from CAD (4.6 g PLA), plus $\frac{1}{2}m_{\text{rod}}L_1^2$.
- **Servo:** the reflected rotor inertia J_s (section 5) dominates the crank inertia by two orders of magnitude.

The plate with its lumped rod masses has mass m , centre of mass $\mathbf{c}$ (plate frame) and inertia $\mathbf{I}_c$ about $\mathbf{c}$. With $\mathbf{r} = \mathbf{R}\mathbf{c}$ and $\mathbf{I}_W = \mathbf{R}\mathbf{I}_c\mathbf{R}^T$, its spatial inertia about $\mathbf{p}$ [11] is

$$\mathbf{H} = \begin{pmatrix} m\mathbf{1} & -m[\mathbf{r}]_{\times} \\ m[\mathbf{r}]_{\times} & \mathbf{I}_W - m[\mathbf{r}]_{\times}[\mathbf{r}]_{\times} \end{pmatrix}, \quad (10)$$

so that the wrench (force, moment about $\mathbf{p}$) needed to give the plate the twist acceleration $\dot{\xi} = (\mathbf{a}_p, \boldsymbol{\alpha}_p)$ is $\mathbf{H}\dot{\xi} + \mathbf{w}_v$ with the velocity-product wrench $\mathbf{w}_v = (m\boldsymbol{\omega}_p \times (\boldsymbol{\omega}_p \times \mathbf{r}), \mathbf{r} \times m\boldsymbol{\omega}_p \times (\boldsymbol{\omega}_p \times \mathbf{r}) + \boldsymbol{\omega}_p \times \mathbf{I}_W\boldsymbol{\omega}_p)$.

4.2 Equations of motion by virtual work

The crank angles are independent generalised coordinates. The principle of virtual work [33, 34] with $\delta \mathbf{x} = \mathbf{J}\delta \boldsymbol{\theta}$ gives

$$\mathbf{D}\ddot{\boldsymbol{\theta}} + \mathbf{J}^T(\mathbf{H}\dot{\xi} + \mathbf{w}_v) = \boldsymbol{\tau}_s + \boldsymbol{\tau}_g + \boldsymbol{\tau}_{\text{stop}} + \mathbf{J}^T\mathbf{w}_{\text{ext}}, \quad (11)$$

with $\mathbf{D} = \text{diag}(J_s + J_{\text{crank}} + \frac{1}{2}m_{\text{rod}}L_1^2)$, servo torques $\boldsymbol{\tau}_s$, crank gravity torques $\tau_{g,i} = -m_{\text{crank}}g d_c \cos(\theta_i + \delta_c) - \frac{1}{2}m_{\text{rod}}gL_1 \cos \theta_i$ (d_c, δ_c locate the crank's centre of mass, CAD), and the external wrench on the plate $\mathbf{w}_{\text{ext}}$: gravity $m\mathbf{g}$ at the centre of mass plus the reaction of the ball, $-\mathbf{F}$ at the contact point and the couple $-\mathbf{M}$ (section 6). Since $\dot{\xi} = \mathbf{J}\ddot{\boldsymbol{\theta}} + \boldsymbol{\kappa}$ with $\boldsymbol{\kappa} = \dot{\mathbf{J}}\dot{\boldsymbol{\theta}}$,

$$(\mathbf{D} + \mathbf{J}^T\mathbf{H}\mathbf{J})\ddot{\boldsymbol{\theta}} = \boldsymbol{\tau}_s + \boldsymbol{\tau}_g + \boldsymbol{\tau}_{\text{stop}} + \mathbf{J}^T(\mathbf{w}_{\text{ext}} - \mathbf{H}\boldsymbol{\kappa} - \mathbf{w}_v). \quad (12)$$

The velocity-product term $\boldsymbol{\kappa}$ is the directional derivative of $\mathbf{J}$ along $\dot{\boldsymbol{\theta}}$, evaluated by a central difference, $\boldsymbol{\kappa} \approx \|\dot{\boldsymbol{\theta}}\| [\mathbf{J}(\boldsymbol{\theta} + \epsilon\dot{\boldsymbol{\theta}}) - \mathbf{J}(\boldsymbol{\theta} - \epsilon\dot{\boldsymbol{\theta}})] / (2\epsilon)$ with $\dot{\boldsymbol{\theta}} = \dot{\boldsymbol{\theta}} / \|\dot{\boldsymbol{\theta}}\|$, $\epsilon = 10^{-6}$; the truncation error is $O(\epsilon^2)$. The plate's acceleration, needed by the ball model, follows from $\dot{\xi}$.

End stops are modelled as a stiff spring-damper outside $[-75^\circ, 75^\circ]$ (AS).

Ball-mechanism coupling. The ball's reaction enters (12) with the value from the previous time step (0.25 ms earlier). The ball's mass is 16% of the plate's and its reaction is a slowly varying force, so this one-step lag is far below every other uncertainty in the model.

5 Servo actuators

The servos are modelled after an MG996R-class metal-gear hobby servo at 6 V [32] (AS choice of servo; see table 3).

Command path. The ESP32-S3 generates the PWM signal with its LEDC peripheral [10]; at 50 Hz and 14-bit resolution the pulse width is quantised to $\Delta = 20 \text{ ms}/2^{14} = 1.22 \mu\text{s}$. A pulse starts at a frame boundary and the servo knows its width only when it ends, so a new command becomes the servo's target w after the frame boundary. The target crank angle is

$$\theta^* = \theta_c + s\pi \frac{w - 1500 \mu\text{s}}{w_{\text{max}} - w_{\text{min}}}, \quad (13)$$

with $[w_{\text{min}}, w_{\text{max}}] = [500, 2500] \mu\text{s}$ spanning 180° , mounting offset θ_c and direction $s = \pm 1$.

Internal controller. Hobby servos close a proportional loop on a potentiometer. It is sampled at $f_c = 250$ Hz (AS) with zero-order hold, the measurement carries Gaussian noise (0.05° , AS), and the drive level is

$$u = \begin{cases} 0, & |e| < \frac{1}{2}\Delta_{\text{db}} \\ \text{sat}_{[-1,1]}(e/e_{\text{pb}}), & \text{otherwise} \end{cases}, \quad e = \theta^* - \theta_{\text{meas}}, \quad (14)$$

with deadband $\Delta_{\text{db}} = 5 \mu\text{s} \hat{=} 0.45^\circ$ (DS) and proportional band $e_{\text{pb}} = 6^\circ$ (AS).

Motor and gearbox. A permanent-magnet DC motor with negligible inductance driven at a fraction u of the supply has a linear torque–speed characteristic [13]. Referred to the output shaft,

$$\tau_s = \tau_{\text{stall}} u - \frac{\tau_{\text{stall}}}{\omega_0} \dot{\theta} - \tau_c \tanh(\dot{\theta}/\omega_\epsilon), \quad (15)$$

with stall torque $\tau_{\text{stall}} = 1.08$ N m and no-load speed $\omega_0 = 7.48$ rad/s (DS), and a regularised Coulomb gear friction τ_c (AS). At $u = 0$ the bridge short-circuits the motor, which brakes it. The reflected inertia is set through the mechanical time constant $\tau_m = J_s \omega_0 / \tau_{\text{stall}} = 20$ ms (AS), giving $J_s = 2.9 \times 10^{-3}$ kg m².

6 Ball–plate dynamics

This is the heart of the model. The ball is a rigid sphere of radius a and mass m_b in contact with the plate, whose motion—pose, twist and twist acceleration—comes from section 4. The classical treatments of a sphere rolling on a moving surface are Neimark and Fufaev [26] and Goldstein et al. [14]. Ball-on-plate control studies usually linearise the result to $\ddot{x} = \frac{5}{7}g s_x$ [2], as is done for the ball and beam [19]. The simulator keeps the full three-dimensional, nonlinear form.

6.1 Rolling inertia

The moment of inertia is $I = k m_b a^2$. For a sphere with a concentric cavity of radius b (a solid ball has $b = 0$),

$$k = \frac{2}{5} \frac{a^5 - b^5}{a^3(a^3 - b^3)}, \quad k = \frac{2}{5} \text{ (solid)}, \quad k \rightarrow \frac{2}{3} \text{ (thin shell)}. \quad (16)$$

As shown below, a rolling ball responds to tangential forces as if its mass were $m_b(1 + k)$: a solid ball gets $\frac{1}{1+k} = \frac{5}{7}$ of the acceleration a sliding particle would, and a table-tennis ball only $\frac{3}{5}$. The default ball is solid bearing steel ($m_b = 32.7$ g, AS material).

6.2 Plate motion as a velocity field

For a plate with origin velocity $\mathbf{v}_p$, acceleration $\mathbf{a}_p$, angular velocity $\boldsymbol{\omega}_p$ and angular acceleration $\boldsymbol{\alpha}_p$, the velocity and acceleration of the plate’s material point at $\mathbf{x}$ are

$$\mathbf{u}(\mathbf{x}) = \mathbf{v}_p + \boldsymbol{\omega}_p \times (\mathbf{x} - \mathbf{p}), \quad (17)$$

$$\mathbf{a}_u(\mathbf{x}) = \mathbf{a}_p + \boldsymbol{\alpha}_p \times (\mathbf{x} - \mathbf{p}) + \boldsymbol{\omega}_p \times (\boldsymbol{\omega}_p \times (\mathbf{x} - \mathbf{p})), \quad (18)$$

and the normal turns as $\dot{\mathbf{n}} = \boldsymbol{\omega}_p \times \mathbf{n}$.

6.3 Newton–Euler equations

With centre $\mathbf{r}$, velocity $\mathbf{v}$ and angular velocity $\boldsymbol{\omega}$, the contact point is $\mathbf{c} = \mathbf{r} - a\mathbf{n}$. Let $\mathbf{F}$ be the total contact force on the ball (normal plus friction) and $\mathbf{M}$ the contact couple (rolling and pivoting resistance, below). Then

$$m_b \dot{\mathbf{v}} = m_b \tilde{\mathbf{g}} + \mathbf{F}, \quad \tilde{\mathbf{g}} = \mathbf{g} - \frac{\rho_{\text{air}} C_d \pi a^2}{2m_b} \|\mathbf{v}\| \mathbf{v}, \quad (19)$$

$$I \dot{\boldsymbol{\omega}} = -a \mathbf{n} \times \mathbf{F} + \mathbf{M}. \quad (20)$$

Air drag is included for completeness. At 0.5 m/s it is 7×10^{-5} of the ball’s weight.

6.4 Rolling without slipping

The contact point of the ball moves with the plate:

$$\mathbf{v} - a\boldsymbol{\omega} \times \mathbf{n} = \mathbf{u}(\mathbf{c}). \quad (21)$$

This single vector equation contains both the persistence of contact (normal component) and no-slip (tangential components). Differentiating it in time, using $\dot{\mathbf{c}} = \mathbf{v} - a\dot{\mathbf{n}}$ and $\frac{d}{dt}\mathbf{u}(\mathbf{c}) = \mathbf{a}_u(\mathbf{c}) + \boldsymbol{\omega}_p \times (\dot{\mathbf{c}} - \mathbf{u}(\mathbf{c}))$, gives

$$\dot{\mathbf{v}} - a\dot{\boldsymbol{\omega}} \times \mathbf{n} = \mathbf{b}, \quad \mathbf{b} = \mathbf{a}_u(\mathbf{c}) + \boldsymbol{\omega}_p \times (\dot{\mathbf{c}} - \mathbf{u}(\mathbf{c})) + a\boldsymbol{\omega} \times (\boldsymbol{\omega}_p \times \mathbf{n}). \quad (22)$$

From (20), $\dot{\boldsymbol{\omega}} \times \mathbf{n} = \frac{1}{I}(-a\mathbf{F}_t + \mathbf{M} \times \mathbf{n})$. Substituting this and (19) into (22) and splitting it into normal and tangential parts gives the accelerations explicitly:

$$\dot{v}_n = b_n, \quad N = m_b(b_n - \tilde{\mathbf{g}} \cdot \mathbf{n}), \quad (23)$$

$$\dot{\mathbf{v}}_t = \frac{k\mathbf{b}_t + \tilde{\mathbf{g}}_t + \frac{\mathbf{M} \times \mathbf{n}}{m_b a}}{1 + k}, \quad \mathbf{F} = m_b(\dot{\mathbf{v}} - \tilde{\mathbf{g}}), \quad (24)$$

and $\dot{\boldsymbol{\omega}}$ from (20). Equation (24) is the rolling-inertia law in its general form. Two special cases are worth stating, because they are the verification cases of section 9:

- *Fixed incline* of slope β ($\mathbf{b} = \mathbf{0}$, $\mathbf{M} = \mathbf{0}$): $\|\dot{\mathbf{v}}\| = \frac{g \sin \beta}{1 + k} = \frac{5}{7}g \sin \beta$ for a solid sphere.
- *Turntable* spinning at constant $\boldsymbol{\Omega} \parallel \mathbf{n}$: here $\mathbf{b} = \boldsymbol{\Omega} \times \mathbf{v}$, so $\dot{\mathbf{v}} = \frac{k}{1+k}\boldsymbol{\Omega} \times \mathbf{v}$. The velocity vector turns at $\frac{k}{1+k}\boldsymbol{\Omega} = \frac{2}{7}\boldsymbol{\Omega}$ and the ball circles at that rate in the inertial frame, whatever its initial condition [35].

The second case shows why the full model matters on a moving plate: the term $k\mathbf{b}_t$ couples the plate's rotation into the ball's motion through the ball's spin, an effect that the point-mass and small-angle models leave out entirely.

6.5 Unilateral contact and friction

Equation (23) gives the normal force needed to keep contact. If $N < 0$, for example when the plate drops faster than g , the contact opens and the ball enters free flight. Rolling requires the friction force to stay inside the Coulomb cone, $\|\mathbf{F}_t\| \leq \mu_s N$ [21, 28]; otherwise the ball slips.

Slipping. While the slip velocity $\mathbf{s} = (\mathbf{v} - a\boldsymbol{\omega} \times \mathbf{n} - \mathbf{u}(\mathbf{c}))_t$ is non-zero, kinetic friction $\mathbf{F}_t = -\mu_k N \mathbf{s}/\|\mathbf{s}\|$ acts. The normal force follows from the second derivative of the gap $d = (\mathbf{r} - \mathbf{p}) \cdot \mathbf{n} - a$:

$$\dot{\mathbf{v}} \cdot \mathbf{n} = \mathbf{a}_p \cdot \mathbf{n} - 2(\mathbf{v} - \mathbf{v}_p) \cdot \dot{\mathbf{n}} - (\mathbf{r} - \mathbf{p}) \cdot (\boldsymbol{\alpha}_p \times \mathbf{n} + \boldsymbol{\omega}_p \times \dot{\mathbf{n}}), \quad (25)$$

which yields N through (19). Slipping ends when the slip velocity vanishes or reverses within a step. The residual slip is then removed by a tangential impulse $\mathbf{P} = -m_b \frac{k}{1+k} \mathbf{s}$ at the contact point. Because this impulse conserves angular momentum about the contact point, it reproduces the exact rolling speed. For a ball launched sliding at v_0 without spin on a level plate, rolling begins at $t^* = \frac{k v_0}{(1+k)\mu_k g}$ at speed $\frac{v_0}{1+k} = \frac{5}{7}v_0$.

Rolling resistance. Hysteresis and micro-slip in the contact produce a couple that opposes rolling [21]. It is modelled as

$$\mathbf{M}_{rr} = -c_{rr} a N \frac{\boldsymbol{\omega}_{\text{rel},t}}{\|\boldsymbol{\omega}_{\text{rel},t}\|}, \quad \boldsymbol{\omega}_{\text{rel}} = \boldsymbol{\omega} - \boldsymbol{\omega}_p, \quad (26)$$

so a ball rolling on a level plate decelerates at $c_{rr}g/(1+k)$. Near zero relative rolling speed the model switches to the stick-slip scheme of Karnopp [22]. Once the relative speed is below 10^{-4} m/s, the ball is *stuck*: it moves rigidly with the plate for as long as the couple needed to hold it satisfies $\|\mathbf{M}_{\text{req},t}\| \leq c_{rr} a N$. On a static incline this holds while $\tan \beta \leq c_{rr}$. With $c_{rr} = 10^{-3}$ the platform therefore has a physical dead zone of 0.057° , below which tilting does not start the ball. This is a real non-linearity that a controller encounters.

Pivoting (drilling) friction. Spin about the normal is resisted by friction over the Hertzian contact patch. For a Hertz pressure distribution of radius $a_c = (3NR/4E^*)^{1/3}$, the limiting couple is $M_p = \frac{3\pi}{16}\mu_k N a_c$ [21], where $1/E^* = (1 - \nu_1^2)/E_1 + (1 - \nu_2^2)/E_2$. For the steel ball on PMMA, $E^* = 3.65$ GPa, $a_c = 87 \mu\text{m}$ and $M_p = 6.6 \times 10^{-6}$ N m. The same Karnopp scheme is used: the relative spin is held at zero while the required couple is below M_p .

6.6 Flight, impact and the plate edge

In flight, $\dot{\mathbf{v}} = \tilde{\mathbf{g}}$ and $\dot{\boldsymbol{\omega}} = \mathbf{0}$. When the ball reaches the plate within its radius while approaching it ($v_n = (\mathbf{v} - \mathbf{u}(\mathbf{c})) \cdot \mathbf{n} < 0$), an impact is resolved by impulses [31]. The normal impulse is $P_n = -m_b(1 + e)v_n$ (Newton’s restitution law; $e = 0$ below a resting speed of 2 cm/s). The tangential impulse is the one that would stop the slip, $-m_b \frac{k}{1+k} \mathbf{s}$, limited to $\mu_k P_n$. Contact resumes, rolling or slipping, once the rebound is below the resting speed. If the contact point leaves the plate disc, the ball goes into flight and is declared lost once its centre is below the plate surface outside the disc.

6.7 Summary of contact modes

Mode	Dynamics	Leaves the mode when
Rolling	(23)–(24)	$N < 0$ (flight); $\ \mathbf{F}_t\ > \mu_s N$ (slip); relative speed $< 10^{-4}$ m/s and holding couple available (stuck)
Stuck	rigid motion with the plate	$N < 0$; friction or rolling-resistance limit exceeded
Slipping	Coulomb friction	slip reverses or vanishes (rolling); $N < 0$
Flight	free fall with drag	impact on the plate; below the plate outside the disc (lost)

7 Camera and image formation

7.1 Projection

The camera looks along $+\mathbf{e}_z$ from $\mathbf{C}$ (table 1). It uses the pinhole model [18] with Brown–Conrady distortion [7], the model used by standard calibration methods [37]. Normalised coordinates (x, y) map to pixels as

$$\begin{aligned} x_d &= x(1 + k_1 r^2 + k_2 r^4 + k_3 r^6) + 2p_1 xy + p_2(r^2 + 2x^2), \\ y_d &= y(1 + k_1 r^2 + k_2 r^4 + k_3 r^6) + p_1(r^2 + 2y^2) + 2p_2 xy, \end{aligned} \quad (u, v) = (f x_d + c_x, f y_d + c_y), \quad (27)$$

with $r^2 = x^2 + y^2$. Pixel (i, j) covers $[i, i + 1) \times [j, j + 1)$. The default image is 320×240 (QVGA). The focal length is set by a 90° horizontal field of view of the undistorted pinhole model ($f = 160$ px), and barrel distortion $k_1 = -0.25$, $k_2 = 0.05$ (AS) widens the actual field to 110° . The inverse mapping uses fixed-point iteration. The ball then appears about 22 px across.

7.2 Refraction through the plate

The camera sees the ball through the 5 mm PMMA plate ($n = 1.49$). At each face the ray refracts by Snell’s law [20], written in vector form for a unit direction $\mathbf{d}$, a surface normal $\mathbf{m}$ facing the ray and $\eta = n_1/n_2$:

$$\mathbf{d}' = \eta \mathbf{d} + \left(\eta \cos i - \sqrt{1 - \eta^2(1 - \cos^2 i)} \right) \mathbf{m}, \quad \cos i = -\mathbf{m} \cdot \mathbf{d}. \quad (28)$$

A plane-parallel slab leaves the direction of the ray unchanged but shifts it sideways by

$$\delta = t_p \frac{\sin(i - r)}{\cos r}, \quad \sin i = n \sin r. \quad (29)$$

At the plate edge ($i \approx 37^\circ$) this is 1.2 mm. A vision pipeline that ignored refraction would make an error of that size, which is larger than the tracking resolution. The firmware and the simulator share the same back-projection routine. It traces the pixel ray through both faces and intersects it with the plane a above the plate top, where the ball centre lies.

7.3 Radiometry and shading

Radiances are in relative units, with the ceiling at 1.

- **Ceiling** (seen through the plate): uniform radiance with a linear gradient of 15% across the field, representing a window (AS). Directions below the horizon see the dark base (0.05).

- **Plate transmittance:** unpolarised Fresnel reflectance $R(i)$ at each face [20], an internal transmittance $\tau = e^{-\alpha t_p / \cos r}$, and the incoherent sum over internal reflections of the slab, $T = (1 - R)^2 \tau / (1 - R^2 \tau^2)$. This gives $T = 0.92$ at normal incidence.
- **Joint bosses:** opaque discs of radius 5 mm under the plate at the joint positions.
- **Ball:** Lambertian diffuse reflection with hemisphere lighting. For a uniform upper (radiance $L_\uparrow$) and lower ($L_\downarrow$) hemisphere, the irradiance on a surface with normal $\mathbf{N}$ is $E = \pi [L_\uparrow \frac{1+N_z}{2} + L_\downarrow \frac{1-N_z}{2}]$ [27]. A mirror term samples the environment in the reflected direction. An optional ring light at the camera adds an energy-normalised Blinn–Phong highlight [4]. Polished steel seen from below is therefore dark, reflecting the base, with a brighter rim.

Rays are traced through the scene in the manner of Whitted [36], without secondary bounces.

7.4 Rolling shutter and motion blur

Row j of a frame whose first row starts exposing at t_0 integrates light over $[t_0 + j t_{\text{row}}, t_0 + j t_{\text{row}} + t_{\text{exp}}]$, with $t_{\text{row}} = t_{\text{readout}}/H$ [24]. The integral over the exposure and the pixel area is estimated by stratified sampling: $S \times S$ positions per pixel and S_t instants per row. This is distributed ray tracing [9], and it produces anti-aliased edges, motion blur and the skew that a rolling shutter gives a moving ball. The plate pose and ball position at each instant are interpolated from the recorded simulation history: linearly for positions, and by normalised quaternion interpolation for orientation. The frame is handed to the application t_{lat} after its last row finishes. The defaults are 30 fps, 6 ms exposure, 15 ms readout and 3 ms latency (AS).

7.5 Sensor model

The mean photo-electron count of a pixel channel is $\mu = \bar{L} V Q$, where $\bar{L}$ is the sampled mean radiance, $V = \cos^n \varphi$ the relative illumination at field angle φ (vignetting, $n = 2$, AS) and Q scales unit radiance to 70% of the full well at the reference exposure. The recorded signal follows the Poisson–Gaussian model of Foi et al. [12]: shot noise (Gaussian with variance μ) plus read noise $\sigma_r = 4e^-$, clipped at the full well of $6000e^-$ (AS). It is then gamma-encoded ($\gamma = 2.2$), quantised to 8 bits, and delivered as the ESP32 camera driver would: RGB565 in big-endian byte order, or 8-bit luma (BT.601 weights). RGB565 keeps only 5–6 bits per channel, and the resulting banding of smooth gradients is visible in fig. 6.

8 Numerical methods

Integration. The coupled state (crank angles and rates, ball position, velocity and angular velocity) is advanced with the classical fourth-order Runge–Kutta method [16]. The fixed step is $h = 0.25$ ms. The fastest continuous dynamics are the end stops ($\sqrt{k_{\text{stop}}/J_s} \approx 260$ rad/s, so $h\lambda \approx 0.07$), well inside the RK4 stability region. Discrete events (PWM frames, controller samples, camera rows) are handled at step boundaries.

Hybrid contact. The contact mode is held fixed during a step and transitions are detected at its end, a time-stepping treatment of the non-smooth dynamics [1]. Transition times are therefore resolved to one step. This is the only discretisation error that appears in the verification (table 2).

Constraint projection. After each step the ball is projected back onto the active constraints [17]. Its position is moved onto the plate surface and its relative normal velocity removed. In rolling, any residual slip is removed by the momentum-consistent impulse of section 6. This replaces Baumgarte-type stabilisation [3] and keeps the drift at round-off level.

Reproducibility. All noise is drawn from a counter-based generator [29]: a SplitMix64 mix [30] of (seed, stream, index), with Box–Muller normals [6]. Every pixel’s noise depends only on the seed, frame and pixel, so runs are bit-identical across compilers and thread counts. The standard library’s distributions are not, because they are implementation-defined.

9 Verification

Each sub-model is checked against an independent closed-form result by the automated test suite. Table 2 summarises the outcomes, and figs. 2 to 4 show the data.

Table 2: Verification against closed-form results.

Check	Reference result	Agreement
CAD pose reproduced by forward kinematics	plate top at 5.000 mm	$2.6 \mu\text{m}$ (idealisation)
Inverse/forward round trip, 504 poses	identity	$2 \times 10^{-11} \text{ m}$, $3 \times 10^{-10} \text{ rad}$
Torsion over the workspace	$\sigma = 0$, (6)	$9 \times 10^{-17} \text{ rad}$
Parasitic translation	(6)	$6 \times 10^{-18} \text{ m}$
Velocity Jacobian	finite differences	3×10^{-11}
Single- vs. double-precision IK (firmware)	—	$6.8 \times 10^{-7} \text{ rad}$
Rolling on 5° incline, solid / shell ($k = 0.604$)	$g \sin \beta / (1 + k)$	$2 \times 10^{-13} / 1 \times 10^{-13} \text{ rel.}$
Energy, rolling on 8° incline, 1 s	conserved	8×10^{-13} of $m_b g \cdot 1 \text{ cm}$
Sliding $\rightarrow$ rolling	$v^* = v_0 / (1 + k)$, t^*	$1 \times 10^{-14} \text{ rel.}$; t^* within one step
Rolling resistance	$c_{rr} g / (1 + k)$, then stuck	$7 \times 10^{-13} \text{ rel.}$
Turntable orbit	$\frac{k}{1+k} \Omega = \frac{2}{7} \Omega$	$1.5 \times 10^{-13} \text{ rel.}$; closure 10^{-15} m
Normal force, plate accelerating $\pm \frac{1}{2} g$	$m_b(g + a)$	exact; separation at $-1.5g$
Pivoting friction	$\dot{\omega}_n = -M_p / I$	$2 \times 10^{-13} \text{ rel.}$
Bounce	rebound = $e \times$ impact speed	exact (impact speed within one step)
Lens distortion inverse	identity	$3 \times 10^{-7} \text{ px}$
Refraction through the plate	$t_p \sin(i - r) / \cos r$	10^{-17} m
Projection/back-projection through plate	identity	$1.1 \times 10^{-11} \text{ m}$
Ball centroid in rendered image	projected centre	0.14–0.98 mm (silhouette bias)
Ball on a commanded 3° slope (full system)	$\frac{5}{7} g \sin s_x$	0.5%

10 End-to-end demonstration

To exercise the whole chain, the closed loop is run through the simulated camera. A simple dark-blob tracker (threshold, boss masking, intensity-weighted centroid) back-projects the centroid through the plate. A PD law with a filtered finite-difference velocity estimate computes the slope angles, and the shared inverse kinematics turns them into servo pulses. This is the kind of program a student writes. Figure 5 shows the ball following setpoint steps; fig. 6 shows the images the tracker receives. The simulation runs about $2.7\times$ faster than real time with full image synthesis on a desktop PC.

11 Assumptions and limitations

The following are *not* modelled. Each is a known difference between simulator and platform.

- **Mechanism:** gear backlash and joint clearances; friction in the passive joints; rotational inertia of the rods; flexibility of the plate, rods and base; manufacturing tolerances (all legs are identical and ideally placed).
- **Servos:** current limiting and supply-voltage sag under load; temperature dependence; the detailed internal control law of a particular servo model.
- **Ball:** departures from sphericity, surface dirt, and variation of friction over the plate; the ball’s orientation is not tracked (its appearance is rotation-invariant).
- **Camera:** Bayer mosaic and demosaicing, colour response, auto-exposure, lens chromatic aberration and flare; multiple scattering in the plate; rods or base entering the field of view; the specific sensor—the CAD shows an OV2710 module, whereas the ESP32 camera interface supports OV2640/OV5640-class sensors, so the camera parameters are generic.
- **Contact:** the ball–plate contact is rigid (no compliance), and impacts are instantaneous.

Parameters tagged AS should be identified on the real platform. Suggested experiments are: a coast-down on a level plate (c_{rr}); the slope at which a resting ball begins to roll (dead zone, c_{rr}); servo step responses with a known load (τ_m , e_{pb}); camera calibration with a checkerboard [37] (intrinsics, distortion, pose); and dark frames and flat fields for the noise model [12].

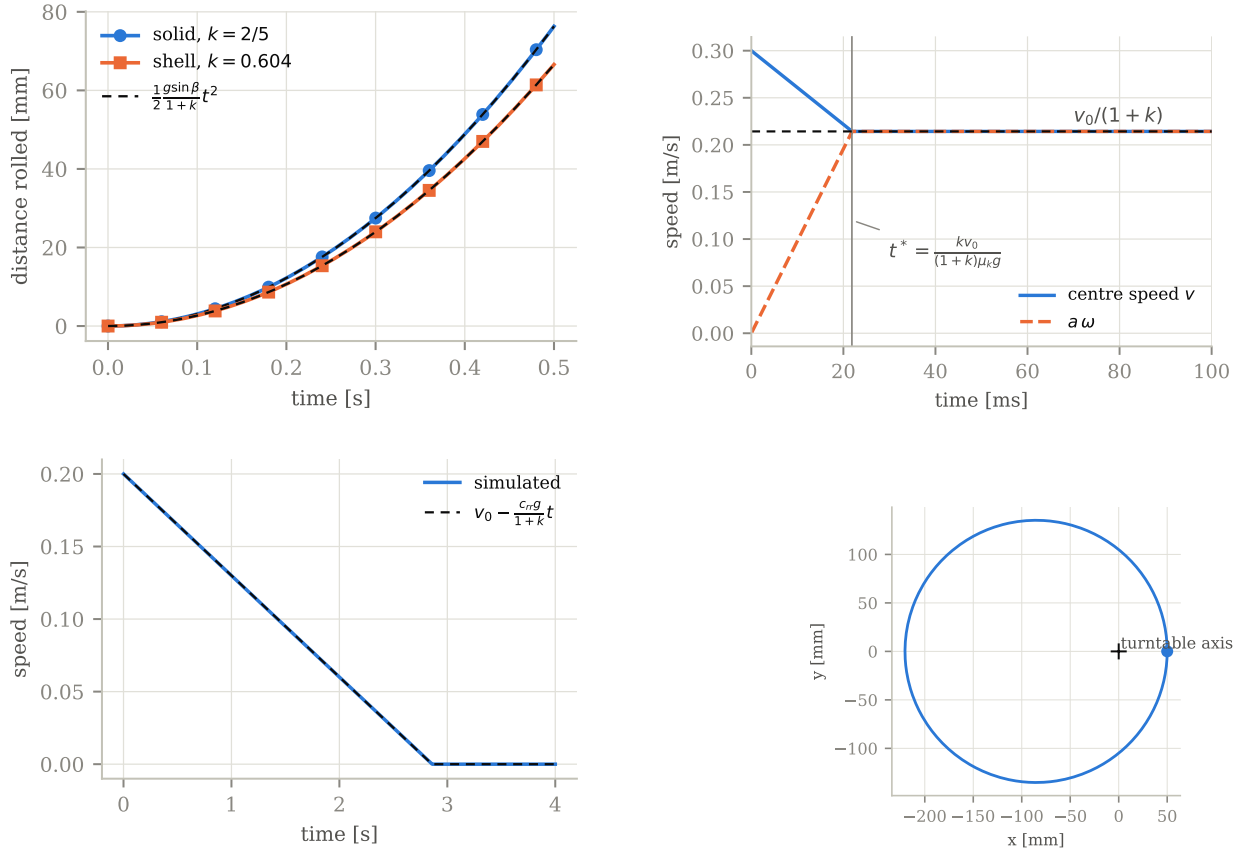


Figure 2: Ball dynamics against theory. Top left: distance rolled down a 5° incline for a solid ball and a hollow shell; the shell’s larger rolling inertia slows it. Top right: a ball launched sliding at 0.3 m/s starts rolling at $\frac{5}{7}v_0$. Bottom left: deceleration under rolling resistance ($c_{rr} = 0.01$), then static hold. Bottom right: path of a ball on a turntable (1 rev/s), one closed orbit every 3.5 turntable revolutions.

A Parameters

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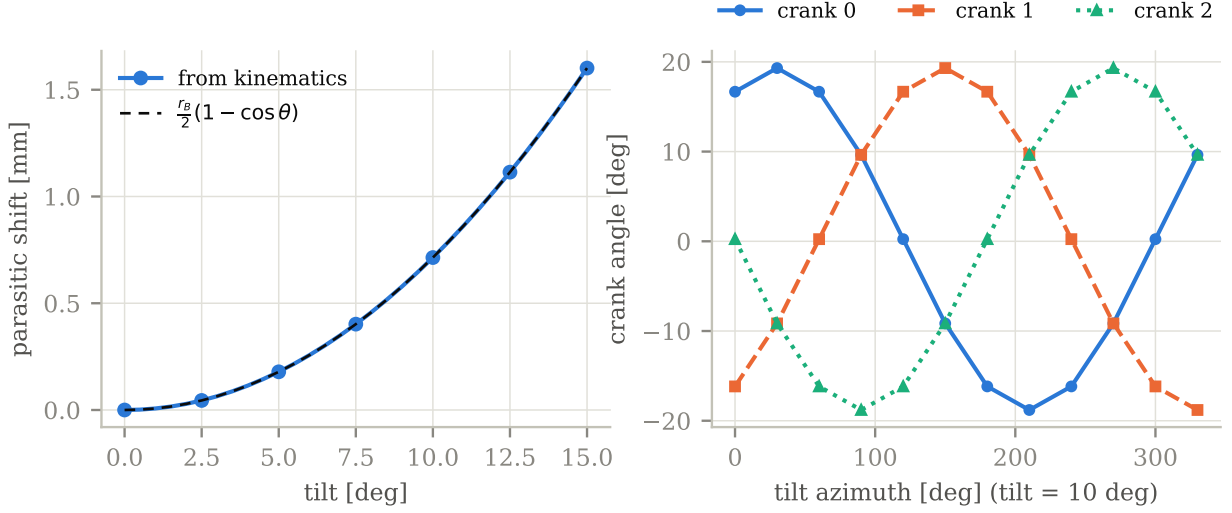


Figure 3: Left: parasitic translation of the joint circle with tilt, against the closed form (6). Right: crank angles for a 10° tilt as its azimuth goes round, the 120° -shifted pattern of the three legs.

Table 3: Physical parameters and their provenance.

Parameter	Value	Source	Note
Gravity	9.80665 m/s ²	standard	
Ball radius / material	10 mm / steel, 7810 kg/m ³	CAD / AS	$m_b = 32.7$ g, $k = 2/5$
Ball E , ν	210 GPa, 0.30	typical steel	Hertz contact
Plate material	PMMA, 1190 kg/m ³	AS	$m_p = 188$ g
Plate E , ν , n	3.2 GPa, 0.37, 1.49	typical PMMA	
Friction μ_s , μ_k	0.45, 0.40	AS	steel on PMMA
Rolling resistance c_{rr}	10^{-3}	AS	dead zone 0.057°
Restitution e	0.5	AS	
Drag coefficient	0.47	sphere	negligible
Crank, rod material	PLA, 1240 kg/m ³ , solid	AS	
Servo stall torque, no-load speed	1.08 N m, 7.48 rad/s	DS MG996R, 6 V	
Servo time constant, friction	20 ms, 0.02 N m	AS	$J_s = 2.9 \times 10^{-3}$ kg m ²
Servo deadband, prop. band, rate	5 μ s, 6° , 250 Hz	DS, AS, AS	
PWM	50 Hz, 1.22 μ s steps, 500–2500 μ s	ESP32-S3 LEDC	
Camera	320 $\times$ 240, HFOV 90° (pinhole), 30 fps	AS	effective 110°
Distortion k_1 , k_2	−0.25, 0.05	AS	
Exposure / readout / latency	6 / 15 / 3 ms	AS	
Full well, read noise	6000 e^- , 4 e^-	AS	
Time step	0.25 ms	numerical	RK4

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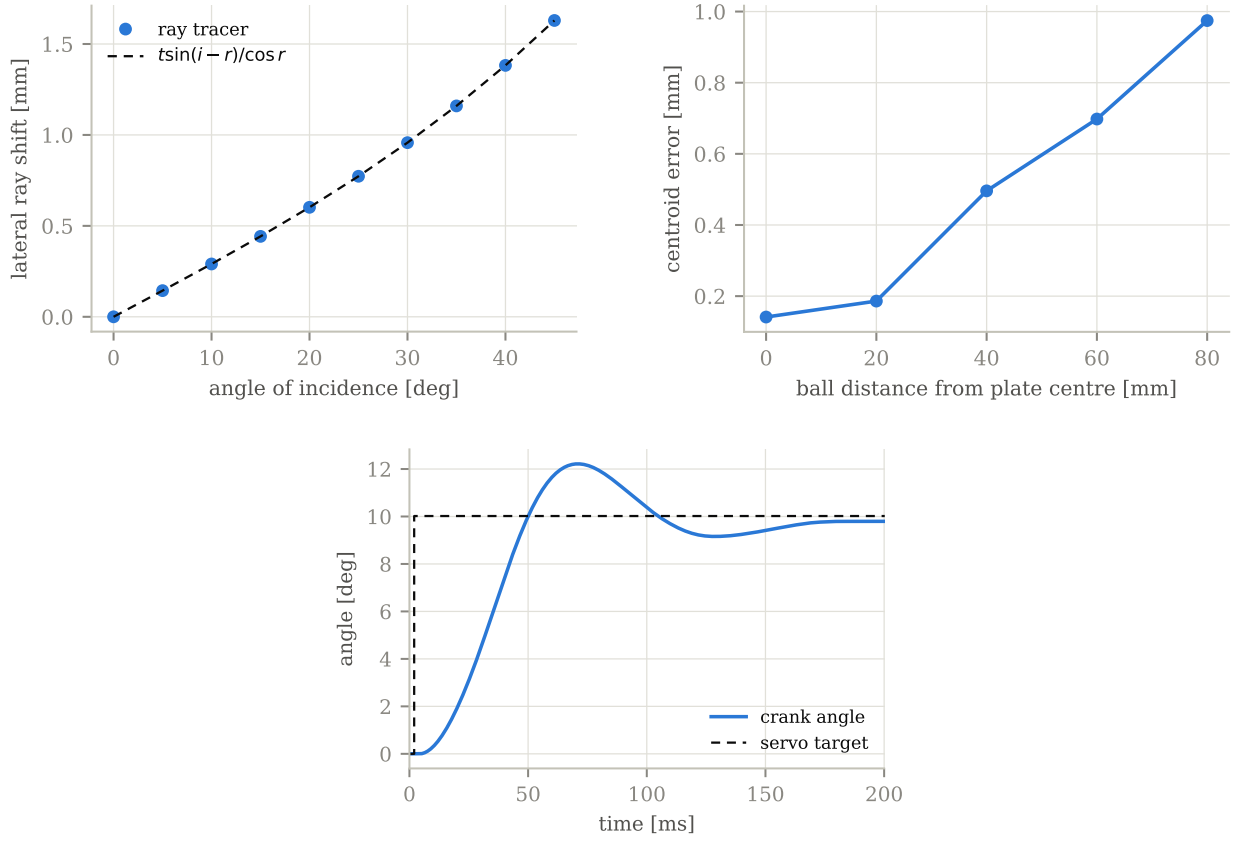


Figure 4: Top left: lateral shift of a camera ray by the plate. Top right: error of the ball position estimated from the intensity-weighted centroid of its silhouette in a noise-free render. Perspective and refraction make the silhouette’s centroid differ from the projection of the centre, which is a systematic error a careful tracker has to account for. Bottom: servo step response (10°) under the platform load. The peak speed is 5.3 rad/s, below the no-load 7.5 rad/s, and the crank settles inside the deadband.

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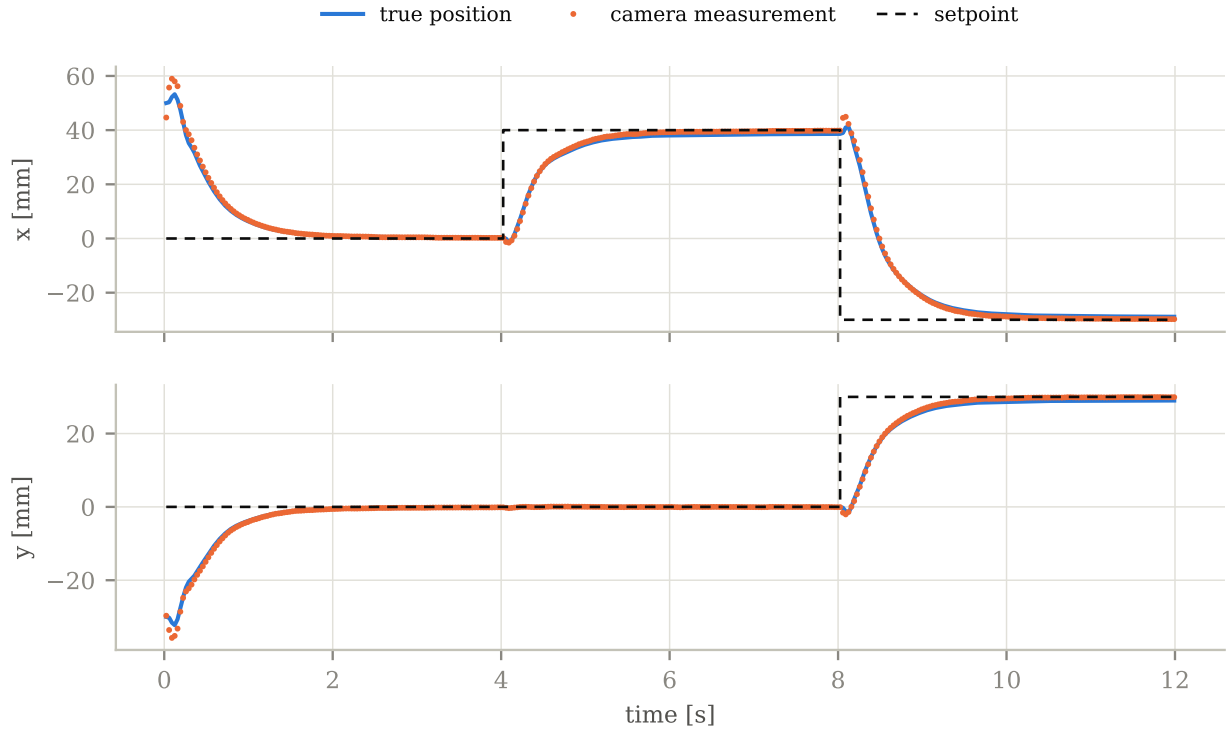


Figure 5: Closed loop through the simulated camera (30 fps) and servos. The ball starts at $(50, -30)$ mm and follows setpoint steps. The final offset of about 1 mm comes from rolling resistance (the dead zone of section 6) together with the centroid bias.

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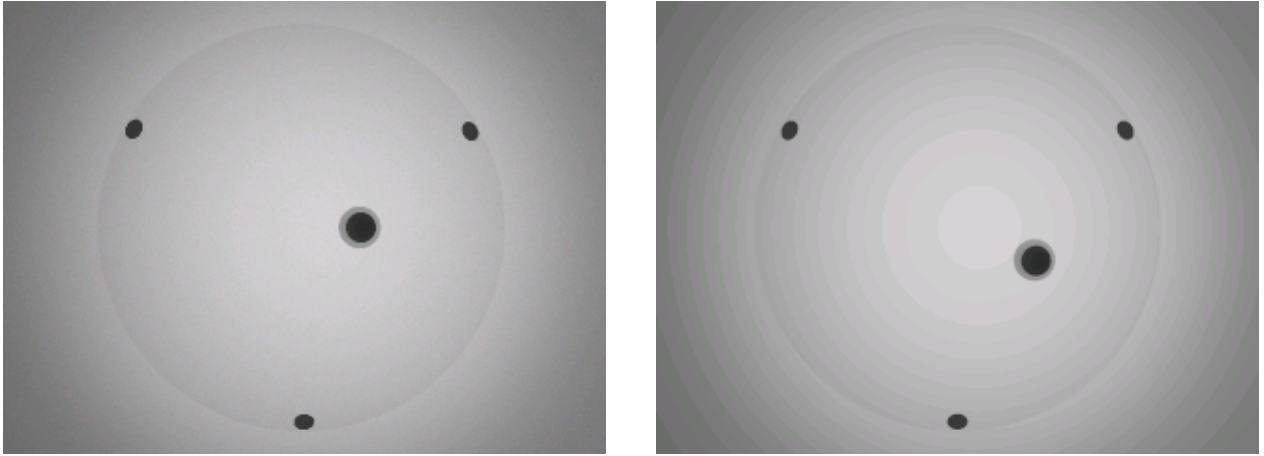


Figure 6: Rendered RGB565 frames. Left: during the closed-loop run, with sensor noise. Right: a noise-free render, where the RGB565 quantisation of the vignetted ceiling shows as rings. The plate disc is slightly darker than its surroundings because of Fresnel losses, and the three joint bosses and the steel ball (dark, with a reflective rim) are visible.